BCS-301	DATA STRUCTURES	L T P 2 1 0	3 credits

Course Objectives:

CO1	Introduce the concept of data structures through ADT including List, Stack, Queues
CO2	To understand concepts about searching and sorting techniques.
CO3	To understand basic concepts about stacks, queues, lists, trees and graphs.
CO4	Able to analyze algorithms and determine their time complexity
CO5	To enable them to write algorithms for solving problems with the help of fundamental data structures.

MODULE-I

Introduction: Basic Terminology, Elementary Data Organization, Algorithm**,** Time and space complexity of algorithms. Asymptotic notations, Abstract data types.
Arrays: Representation of arrays, insertion and deletion operations, Single and Multidimensional Arrays, Sparse Matrices and their representations. **Searching** : Sequential search, Binary Search, Comparison and Analysis
Linked List: Singly linked lists: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from linked list; Linked representation of Stack and Queue, Header nodes, Doubly linked list: operations on it and algorithmic analysis; Circular Linked Lists: all operations their algorithms and the complexity analysis. Garbage collection and memory compaction.

MODULE-II

Stacks: Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack. Application of stack: Prefix and Postfix Expressions conversion, Evaluation of postfix expression, Iteration and Recursion-Principles of recursion, Problem solving using recursion with examples such as binary search, Fibonacci numbers, and towers of Hanoi.
Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues, Dequeue and Priority Queue.

Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List. Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm

MODULE-III

Trees : Basic terminology used with Binary Trees, Binary Tree Representation, Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Binary Search Tree: Operation of Insertation , Deletion, Searching in Binary Search tree . Threaded Binary trees, Traversing Threaded Binary trees..
Concept & Basic Operations of AVL Tree. B Tree: definitions, algorithms and analysis.
Sorting: Bubble sort, Insertion Sort, Selection, Bubble Sort, Quick Sort, Two Way Merge Sort, Heap Sort.

Performance and Comparison among all the methods

Hashing: Hash Function, Collision Resolution Strategies

Text books and References:

1. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein "Data Structures Using C and C++" , PHI

2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publication

3. Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with applications", McGraw Hill

4. R. Kruse etal, "Data Structures and Program Design in C", Pearson Education

5. Lipschutz, "Data Structures" Schaum's Outline Series, TMH

6. G A V Pai, "Data Structures and Algorithms", TMH

COURSE OUTCOMES:

CO1	For a given algorithm student will able to analyze the algorithms to determine the time and computation complexity and justify the correctness.
CO2	For a given Search problem (Linear Search and Binary Search) student will able to implement it.
CO3	For a given problem of Stacks, Queues and linked list student will able to implement it and analyze the same to determine the time and computation complexity.
CO4	Able to analyze algorithms and determine their time complexity
CO5	Student will able to implement Graph search and traversal algorithms and determine the time and computation complexity.